07-06-23

MULTI-DIMENSIONAL ARRAY

Ques

Sample Input

m =

4	5
6	2
9	10

n x 2

column no will always be 2.

Sample Output

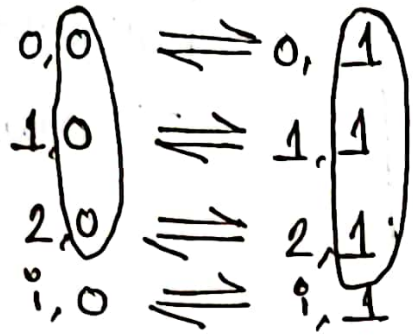
5	4
2	6
10	9

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(0,0) ⇌ (0,1)

P.T.O

Solve



• 2D code → double ordering must!

[swapping code]

```
row, colum = m.shape .
for i in range (row):
    m[i][0], m[i][1] = m[i][1], m[i][0]
```

OR

```
for i ⟶ (row):
    for j ⟶ (col):
        m[i][j] = m[j][i]
```

✗ will not work.

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Ques

Input

4	5	10	11
6	2	1	3
9	10	7	4
0	1	2	3

[swaping two column]

Output

11	10	5	4
3	1	2	6
4	7	10	9

1st col ⇌ 4th col

2nd col ⇌ 3rd col

Col		Col
0	⇌	3
1	⇌	2
i	⇌	col-1-i

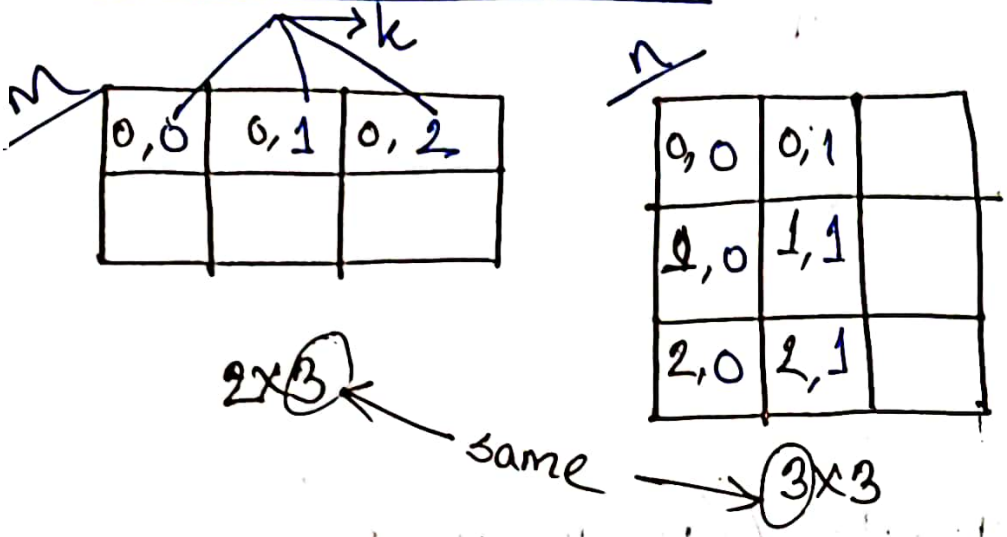
m[0][0] ⇌ m[0][3]

m[0][1] ⇌ m[0][2]

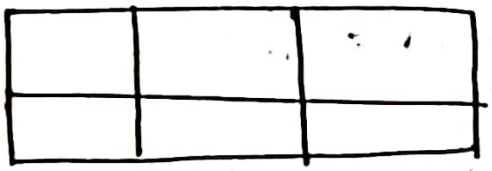
```
for i in range (row):
    for j in ~~range (col)~~: range (col//2):
        m[i][j], m[i][col-1-j] = m[i][col-1-j], m[i][j]
```

Matrix Multiplication:

V.V.I



result



2x3

m-matrix	n-matrix	result
row	col	
0th	0th	[0] [0]
0th	1st	[0] [1]
0th	2nd	[0] [2]
ith	jth	[i] [j]

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```
row_m , col_m = m.shape
row_n , col_n = n.shape.

if col_m != row_n :
    (can't).

else :
    result = np.zeroes((row_m , col_n))

    for i in range (row_m):
        for j in range (col_n):
            for k in range (col_m):
                result [i][j] = result[i][j] +
                m[i][k] * n[k][j]
```

simulation

i	j	k	result
0	[0]	0	row[0][0] = row[0][0] + m[0][0] * n[0][0]
		1	row[0][0] = row[0][0] + m[0][1] * n[1][0]
		2	row[0][0] = row[0][0] + m[0][2] * n[2][0]
[1]	0		row[0][1] = row[0][1] + m[0][0] * n[0][1]
		1	row[0][1] = row[0][1] + m[0][1] * n[1][1]

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